SUPPLEMENTAL MATERIAL

Appendix 1. Search strategy

WEB OF SCIENCE		(Chronic Obstructive Pulmonary Disease OR COPD OR Chronic obstructive lung disease OR Chronic Obstructive Airway Disease OR Chronic Airway Obstruction OR Emphysema OR Chronic bronchitis)
	AND	"step test"
		(Chronic Obstructive Pulmonary Disease OR COPD OR Chronic obstructive lung disease OR Chronic Obstructive Airway Disease OR Chronic Airway Obstruction OR Emphysema OR Chronic bronchitis)
	AND	"stepper test"
		(Chronic Obstructive Pulmonary Disease OR COPD OR Chronic obstructive lung disease OR Chronic Obstructive Airway Disease OR Chronic Airway Obstruction OR Emphysema OR Chronic bronchitis)
	AND	Stepping
MEDLINE		COPD OR chronic obstructive pulmonary disease OR chronic obstructive airway disease OR chronic obstructive lung disease *Select a field
	AND	step test *AB Abstract
		COPD OR chronic obstructive pulmonary disease OR chronic obstructive airway disease OR chronic obstructive lung disease *Select a field
	AND	stepper test *AB Abstract
		COPD OR chronic obstructive pulmonary disease OR chronic obstructive airway disease OR chronic obstructive lung disease *Select a field
	AND	stepping *AB Abstract
PUBMED		("Pulmonary Disease, Chronic Obstructive"[Mesh]) AND "step test"
		("Pulmonary Disease, Chronic Obstructive"[Mesh]) AND "stepper test"
		(("pulmonary disease, chronic obstructive"[MeSH Terms] OR ("pulmonary"[All Fields] AND "disease"[All Fields] AND "chronic"[All Fields] AND "obstructive"[All Fields]) OR "chronic obstructive pulmonary disease"[All Fields] OR "COPD"[All Fields])) AND "Step test"
		(("pulmonary disease, chronic obstructive"[MeSH Terms] OR ("pulmonary"[All Fields] AND "disease"[All Fields] AND "chronic"[All Fields] AND "obstructive"[All Fields]) OR "chronic obstructive pulmonary disease"[All Fields] OR "COPD"[All Fields])) AND "Stepper test"
		(("pulmonary disease, chronic obstructive"[MeSH Terms] OR ("pulmonary"[All Fields] AND "disease"[All Fields] AND "chronic"[All Fields] AND "obstructive"[All Fields]) OR "chronic obstructive pulmonary disease"[All Fields] OR "COPD"[All Fields])) AND (((step) OR (stepper) OR (stepping))

PEDro	AND	COPD “step test”
	AND	COPD "stepper test"
	AND	COPD Stepping
CENTRAL COCHRANE LIBRARY	AND	Pulmonary Disease, Chronic Obstructive *Title Abstract Keyword "step test" *Title Abstract Keyword
	AND	Pulmonary Disease, Chronic Obstructive *Title Abstract Keyword "stepper test" *Title Abstract Keyword
	AND	Pulmonary Disease, Chronic Obstructive *Title Abstract Keyword stepping *Title Abstract Keyword
SCOPUS	AND	"stepper test" *Title Abstract Keyword "step test”
	AND	COPD * Article title Abstract, Keywords "stepper test”
	AND	COPD * Article title, Abstract, Keywords Stepping

Appendix 2. Definition and quality criteria for measurement properties

Domain	Property/Definition	Rating	Quality criteria
Validity	Criterion validity: degree to which the scores of a PROM are an adequate reflection of a gold standard.	+	Correlation with gold standard ≥0.70 OR AUC ≥0.70
		?	Not all information for "+" reported
		-	Correlation with gold standard <0.70 OR AUC <0.70
	Construct validity/hypotheses testing: degree to which the scores of a PROM are consistent with hypotheses based on the assumption that the PROM validly measures the construct to be measured.	+	Correlations with instruments measuring the same construct are ≥0.50.
		?	Solely correlations determined with unrelated constructs
		-	Correlations with instruments measuring the same construct are <0.50.
Reliability	Reliability: degree to which measurement is free from measurement error.	+	ICC/weighted kappa of ≥0.70 or Pearson r of ≥0.80
		?	Neither ICC/weighted kappa nor Pearson r determined
		-	ICC/weighted kappa of <0.70 or Pearson r of <0.80

	Measurement error: systematic and random error of a patient's score that is not attributed to true changes in the construct to be measured	+	SDC or LoA<MIC
		?	MIC not defined
		-	SDC or LoA>MIC
Responsiveness	Responsiveness: ability of a PROM to detect change over time in the construct to be measured.	+	Correlations with changes on instruments measuring the same construct are ≥0.50 or AUC of ≥0.70 and correlations with changes in related constructs are higher than those with unrelated constructs
		?	Solely correlations determined with unrelated constructs
		-	Correlations with changes on instruments measuring the same construct are <0.50 or AUC of <0.70 and correlations with changes in related constructs are higher than those with unrelated constructs
Interpretability	Interpretability is the to which one can assign qualitative meaning – that is, clinical or commonly understood connotations – to a PROM's quantitative scores or change in scores.		

1 PROM: patient-reported outcome measures; AUC: area under the curve; ICC: intraclass correlation coefficient; SDC: smallest detectable change; LoA: limits of agreement;

2 MIC: minimal important change; +: sufficient; –: insufficient; ?: indeterminate.

3

4

Appendix 3. Characteristics of included studies.

Study (year)	Test	Measurement Property Assessed	Description of the Sample		COSMIN box/Domain Assessed	Study Aims	Outcomes measures
			Number of Participants	Age, years Mean (SD)			
de Camargo et al (2011)[31]	CST	Criterion validity	11	No age data available.	Box 8	To determine the reliability of the CST and the correlation with exercise test results (CPET, 6MWT).	CST: number of steps; CPET: W_{peak}; 6MWT: 6MWD.
		Construct validity (hypotheses testing)	32	69.9 (9.0)	Box 9		
		Reliability			Box 6		
		Measurement Error			Box 7		
Karloh et al (2013)[32]	CST	Construct validity (hypotheses testing)	10	64.0 (10.0)	Box 9	To compare the response induced by the CST with the response of the ISWT and 6MWT.	CST: number of steps, level reached; ISWT: level reached; 6MWT: 6MWD.
Dal Corso et al (2013)[33]	MIST	Criterion validity	34	67.0 (9.0)	Box 8	To compare maximal physiological responses	MIST: VO_{2peak}, SpO_2; CPET: VO_{2peak}, SpO_2.
		Reliability			Box 6		

		Measurement Error Interpretability			Box 7	between MIST and a CPET. To test the reproducibility of the MIST on different days.	
de Blok et al (2006)[44]	2MST	Responsiveness	10	65.7 (10.4)	Box 10	To investigate the short-term effects of a lifestyle physical activity counseling program with feedback of a pedometer during a standard pulmonary rehabilitation on daily physical activity, and on physical fitness.	2MST: number of steps
Vieira et al (2020)[54]	4MST	Interpretability	186	65.0 (9.0)		To investigate whether a 4MST would be a suitable surrogate of the 6MWT, in a modified BODE step index to predict mortality.	4MST: number of steps

Marrara et al (2012)[45]	6MST	Responsiveness	36	70.5 (8.5) and 68.3 (8.7)	Box 10	To evaluate the responsiveness of the 6MST to aerobic physical training.	6MST: number of steps, and physiological variables.
Pessoa et al (2014)[34]	6MST	Construct validity (hypotheses testing) Interpretability	32	69.0 (10.0)	Box 9	To compare the 6MST with the 6MWT.	6MST: number of steps; 6MWT: 6MWD.
da Costa et al (2014)[41]	6MST	Reliability Measurement Error	32	68.5 (10.3)	Box 6 Box 7	To evaluate 6MST reliability.	6MST: number of steps.
de Souza et al (2018)[46]	6MST	Responsiveness	50	67.0 (10.0) and 69.0 (11.0)	Box 10	To determine the benefits of PR and whether a rehabilitation manual would facilitate the maintenance of these benefits.	6MST: number of steps.
Munari et al (2020)[42]	6MST	Reliability Measurement Error	36	67.0 (7.0)	Box 6 Box 7	To test the reliability of the 6MST performance.	6MST: number of steps, and

							physiological variables.
Kovelis et al (2020)[47]	6MST	Responsiveness	44	70.0 (8.0) and 70.0 (7.0)	Box 10	To investigate the effectiveness of a supervised home-based physical training program on exercise capacity.	6MST: number of steps
Murphy et al (2005)[48]	3MST	Responsiveness	26	67.0 (9.7) and 65.0 (11.0)	Box 10	To evaluate a simple supervised home-based program initiated immediately after a home from hospital-treated COPD exacerbation, led to improved exercise capacity and reduced exacerbations.	3MST: time, SpO_2, HR and dyspnea (Borg scale 0-10).
Borel et al (2016)[49]	3MST	Responsiveness	40	69.0 (7.0)	Box 10	To evaluate the responsiveness of 3MST	3MST: dyspnea (Borg scale 0-10) and

						(constant) to detect the relief of exertional dyspnea after acute bronchodilation.	physiological variables.
Beaumont et al (2019)[35]	3MST	Construct validity (hypotheses testing)	50	62.3 (7.9)	Box 9	To compare the 3MST with the 6MWT.	3MST: number of steps; 6MWT: 6MWD.
		Reliability			Box 6		
		Measurement Error			Box 7		
Lewthwaite et al (2020)[55]	3MST	Interpretability	98	66.3 (6.6)		To develop COPD-specific equations for predicting the VO_{2peak} during the 3MST.	3MST: VO_{2peak}, and dyspnea (Borg scale 0-10)
Swinburn et al (1985)[29]	Paced Step Test	Criterion Validity	17	66	Box 8	To compare the characteristics of two different types of exercise (step test and a cycloergometer test)	Paced Step Test: number of steps Cycloergometer test: W_{peak}

Perrault et al (2009)[30]	Step test protocol	Reliability	43	65.0 (7.0)	Box 6	To develop a step test that could effectively assess the degree of exertional dyspnea and to demonstrate its reproducibility.	Step test protocol: dyspnea (Borg scale 0-10), HR, VO_{2peak}, and other physiological variables.
		Measurement Error			Box 7		
Coquart et al (2015)[43]	6MSpT	Reliability	35	60.8 (8.9)	Box 6	To test reliability of the 6MSpT, and evaluate its accuracy in detecting improved functional capacity after PR.	6MSpT: number of steps.
		Measurement Error			Box 7		
		Responsiveness			Box 10		
Coquart et al (2016)[50]	6MSpT	Responsiveness	188		Box 10	To compare the effects of home-based PR including NMES and PR with endurance physical exercise on exercise tolerance.	6MSpT: number of steps.
	6MSpT	Criterion validity	91	60.3 (9.3)	Box 8		

Grosbois et al (2016)[36]		Construct validity (hypotheses testing) Responsiveness			Box 9 Box 10	To compare the results of the 6MSpT to those obtained during a 6MWT and CPET.	6MSpT: number of steps; CPET: VO_{2peak}, W_{peak}; 6MWT: 6MWD.
Pichon et al (2016)[37]	6MSpT	Construct validity (hypotheses testing) Responsiveness Interpretability	62	61.8 (9.0)	Box 9 Box 10	To compare the 6MSpT and the 6MWT, to evaluate the responsiveness of the 6MSpT, and to determine a MID.	6MSpT: number of steps; 6MWT: 6MWD.
Bonnevie et al (2017)[38]	6MSpT	Criterion validity	24	61.5 (8.7)	Box 8	To develop a prediction equation to set intensity in patients attending PR, based on the 6MSpT.	6MSpT: number of steps, HR; CPET: HRvt, Wvt.
Fabre et al (2017)[39]	6MSpT	Criterion validity Construct validity (hypotheses testing)	28 50	No age data available. 60.1 (8.5)	Box 8 Box 9	To compare the heart rate (HR) responses to three exercise modalities measuring aerobic fitness	6MSpT: HR; CPET: HR; 6MWT: HR;

Miranda et al (2019)[51]	6MSpT	Responsiveness	21	64.1 (9.9)	Box 10	To evaluate the acute effect of photobiomodulation therapy with combined use of lasers, LEDs, and magnetic field on muscle performance, exercise tolerance, and metabolic variables.	6MSpT: number of steps, SpO_2, VO_{2peak}, HR, and dyspnea/fatigue (Borg scale 0-10).
Bonnevie et al (2019)[40]	6MSpT	Criterion validity	35	61.0 (9.0)	Box 8	To assess whether the 6MSpT could be used to prescribe strength training.	6MSpT: number of steps; Quadriceps muscle: 1RM.
Grosbois et al (2019)[52]*	6MSpT	Responsiveness	459	64.2 (11.3)	Box 10	To determine whether socioeconomic deprivation, affected the short- and long-term outcomes of a home-based PR program.	6MSpT: number of steps.
Bonnevie et al (2020)[53]	6MSpT	Responsiveness	56	62.0 (9.0)	Box 10	To assess the prevalence of cognitive dysfunction among	6MSpT: number of steps.

						people with COPD referred for PR and the mid-term effects of PR.	

CST: Chester Step Test; MIST: Modified Incremental Step Test; 2MST: 2-minute Step Test; 3MST: 3-minute Step Test; 4MST: 4-minute Step Test; 6MST: 6-minute step test; 6MSpT: 6-minute stepper test; CPET: cardiopulmonary exercise test 6MWT: 6-minute walk test; ISWT: incremental shuttle walk test; Wpeak: peak workload; 6MWD: 6-minute walk distance; NMES: transcutaneous neuromuscular electrical stimulation; SpO2: peripheral oxygen saturation; VO2peak: peak oxygen consumption; HR: heart rate; HRvt: heart rate at ventilatory threshold WVT: workload at ventilatory threshold; 1RM: one-repetition maximum; PR: pulmonary rehabilitation.

* The 6-minute Stepper Test was applied in four more studies to verify the benefits of home-based PR programs, but the patients of these studies are included in this sample of participants.

Appendix 4. Criterion validity, construct validity (hypothesis testing), reliability, measurement error and responsiveness of the step tests.

Study (year)	Test	Criterion validity			Construct validity (hypotheses testing)			Responsiveness		
		n	Methodological quality	Result (rating)	n	Methodological quality	Result (rating)	n	Methodological quality	Result (rating)
de Camargo et al (2011)	CST	11	Very good	r= 0.69 (-)	32	Very good	r=0.60 (+)			
Karloh et al (2013)					10	Very good	r= 0.76-0.83 (+)			
Pooled for summary result (overall rating)					**42**		**r=0.60-0.83 (+)**			
Dal Corso et al (2013)	MIST	34	Inadequate (1)	(?)						
de Blok et al (2006)	2MST							10	Inadequate (9, 12)	NA
Pessoa et al (2014)	6MST				32	Very good	r=0.76 (+)			
Marrara et al (2012)								36	Inadequate (9)	NA
de Souza et al (2018)								50	Inadequate (9, 12)	NA

Kovelis et al (2020)								44	Inadequate (9,12)	NA
Murphy et al (2005)	3MST							26	Inadequate (9, 12)	NA
Borel et al (2016)								40	Inadequate (12)	NA
Beaumont et al (2019)					50	Very good	r= 0.78 (+)			
Swinburn et al (1985)	Paced Step Test	17	Very good	r=0.75 (+)						

Appendix 4. (continued)

Study (year)	Test	Reliability			Measurement Error		
		n	Methodological quality	Result (rating)	n	Methodological quality	Result (rating)
Camargo et al (2011)	CST	32	Inadequate (2)	ICC=0.99 (+)	32	Inadequate (2)	Agreement: -20.2 to 17.9 steps (bias of -1.1 steps) (?)

Dal Corso et al (2013)	MIST	34	Very good	ICC=0.93-0.99 (+)	34	Very good	Agreement: -32 to 25 steps (bias of -3.5 steps) and -0.17 to 0.15 L in VO_{2peak} (bias of -0.01L) (?)
da Costa et al (2014)	6MST	32	Inadequate (3)	ICC= 0.94 (+)	32	Inadequate (3)	SEM=4.8 steps; MDD= 11.1 steps; agreement: 7.1 to 18.6 steps (bias of 5.75 steps) (?)
Munari et al (2020)		36	Inadequate (2)	ICC=0.98 (+)	36	Inadequate (2)	SEM=4.27; MDD= 11.8 (?)
Pooled for summary result (overall rating)		**68**		**ICC=0.94-0.98 (+)**	**68**		**SEM=4.27-4.8; MDD= 11.1-11.8 (?)**
Beaumont et al (2019)	3MST	50	Inadequate (2)	ICC=0.96 (+)	50	Inadequate (2)	Agreement: -8.4 to 18.08 steps (bias of 4.84 steps) (?)
Perrault et al (2009)	Step test protocol	43	Very good	ICC>0.91 (+)	43	Very good	Agreement available (?)

CST: Chester Step Test; MIST: Modified Incremental Step Test; 2MST: 2-minute Step Test; 3MST: 3-minute Step Test; 4MST: 4-minute Step Test; 6MST: 6-minute step test; 6MSpT: 6-minute stepper test; +: sufficient; –: insufficient; ?: indeterminate; NA: not applicable; ICC: Intraclass Correlation Coefficient; SEM: Standard Error of Measurement; MDD: Minimal Detectable Difference. On the methodological quality column, the numbers in parentheses indicate the items that justify the score according to the COSMIN risk-of-bias checklist.

Appendix 4. (continued)

Study (year)	Test	Criterion validity			Construct validity (hypotheses testing)			Responsiveness		
		n	Methodological quality	Result (rating)	n	Methodological quality	Result (rating)	n	Methodological quality	Result (rating)
Coquart et al (2015)	6MSpT							35	Inadequate (12)	NA
Coquart et al (2016)								188	Inadequate (9,12)	NA
Grosbois et al (2016)		91	Very good	r=0.36-0.46 (-)	91	Very good	r= 0.56 (+)	91	Very good	r=0.32-0.34 (-)
Pichon et al (2016)					62	Very good	r= 0.71 (+)	62	Very good	r=0.26 (-)
Bonnevie et al (2017)		24	Very good	r= 0.57-0.69 (-)						
Fabre et al (2017)		28	Very good	r= 0.48 (-)	50	Very good	r=0.62-0.68 (+)			
Miranda et al (2019)								21	Inadequate (9, 12)	NA
Bonnevie et al (2019)		35	Very good	r=0.44 (-)						

Grosbois et al (2019)								459	Inadequate (9,12)	NA
Bonnevie et al (2020)								56	Inadequate (9,21)	NA
Pooled for summary result (overall rating)		**143**		**r= 0.36-0.69 (-)** gold standard: CPET	**203**		**r= 0.56-0.71 (+)**	**153**		**r=0.26-0.34 (-)**

Appendix 4. (continued)

Study (year)	Test	Reliability			Measurement Error		
		n	Methodological quality	Result (rating)	n	Methodological quality	Result (rating)
Coquart et al (2015)	6MSpT	35	Inadequate (2)	Baseline: ICC=0.91, r=0.92 (+);	35	Inadequate (2)	Baseline -agreement: -134 to 51 steps (bias of -42 steps); SEM= 37.1-40.4 steps (?);

				Post-PR: ICC=0.94, r=0.95(+)			Post-PR – agreement: 138 to 34 steps (bias of -52 steps; SEM=40.8-47.1 steps. (?)

6MSpT: 6-minute stepper test; +: sufficient; –: insufficient; ?: indeterminate; NA: not applicable; ICC: Intraclass Correlation Coefficient; SEM: Standard Error of Measurement; MDD: Minimal Detectable Difference. On the methodological quality column, the numbers in parentheses indicate the items that justify the score according to the COSMIN risk-of-bias checklist.